SAMARTH BIRAJDAR

Computer Science Engineer | B.Tech - Final Year
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PROFESSIONAL SUMMARY

Motivated Computer Science undergraduate (Final Year, CGPA 7.29) at Pimpri Chinchwad University with hands-on research and project experience in Machine Learning, NLP, Time-Series Forecasting, and Bio-Inspired Optimization. Completed a Software Developer & QA Tester internship at ShieldGuardia Software Private Limited, gaining practical exposure to the full Software Development Life Cycle (SDLC). Built an end-to-end Sales Forecasting and Demand Intelligence system (SARIMA, Prophet, XGBoost) and an ML-based Hospital Mortality Prediction System. Co-authored a research paper on News Category Classification using metaheuristic algorithms achieving 87% accuracy with DE-SVM. Holds industry certifications from Meta, Red Hat, AutomationEdge, and Samsung. Seeking a challenging role in Software Development, Machine Learning, or Data Engineering.

SKILLS

Programming Languages: C, Python, HTML, MySQL
CS Concepts: Data Structures, Algorithms, DBMS, OOP, Machine Learning, NLP
Machine Learning / AI: Random Forest, SVM, XGBoost, TF-IDF, PSO, DE, GA, WOA, GWO, SARIMA, Prophet, K-Means Clustering, Isolation Forest
Software Testing / QA: Manual Testing, Test Case Design, Functional Testing, Regression Testing, Defect Reporting, SDLC
Tools & Technologies: RPA (AutomationEdge), Git/GitHub, Streamlit, IoT (Basic)
Soft Skills: Research, Analytical Thinking, Problem Solving, Team Collaboration
Languages Known: English, Marathi, Hindi

EXPERIENCE

Software Developer & QA Tester | *ShieldGuardia Software Private Limited* **May 2026 - Jul 2026**

- Completed a 2-month Quality Assurance & Web Development internship, gaining hands-on exposure to the full Software Development Life Cycle (SDLC) - requirement analysis, design, development, testing, and deployment.
- Contributed to web application development and manual QA testing, including test case preparation, functional and regression testing, and defect identification and reporting to improve software quality.
- Collaborated with the project manager and development team following structured SDLC practices; awarded an overall performance rating of Excellent.

Certificate of Internship Completion No. SG-2026-008, issued 19 July 2026

EDUCATION

B.Tech, Computer Science | *Pimpri Chinchwad University, Pune* **2023 - Present**

CGPA: 7.29

HSC | *Mahatma Gandhi Global Village, Boramni* **2022 - 2023**

Percentage: 74.4%

SSC | *St. Joseph High School, Solapur* **2020 - 2021**

Percentage: 71%

PROJECTS

Sales Forecasting & Demand Intelligence | Python, Machine Learning, Time-Series Forecasting

GitHub | Live Streamlit App | Analysis Notebook

- Analyzed 4 years (2015-2018) of retail sales transactions (9,800 orders, $2.26M total revenue) to build a full sales forecasting and demand intelligence pipeline.
- Built and benchmarked three forecasting models - SARIMA, Prophet, and XGBoost; XGBoost achieved the lowest error (13.4% MAPE) and was selected to project Q1 2019 revenue.
- Applied Isolation Forest and Z-Score anomaly detection techniques to flag 11 statistically unusual weekly sales patterns for business investigation.
- Used K-Means clustering to segment 17 product sub-categories into four demand profiles, translating raw sales data into actionable inventory strategies.
- Deployed an interactive Streamlit web application to visualize forecasts, anomalies, and demand segments for business stakeholders.

Hospital Mortality Rate Prediction System | Python, Machine Learning, EHR Data

Guide: Dr. Sachin Jadhav | Pimpri Chinchwad University

- Built an ML system to predict in-hospital patient mortality using EHR data including vital signs, lab results, comorbidity indexes, and patient demographics.
- Implemented Random Forest, SVM, and Neural Networks with a Decision Fusion layer to improve overall prediction accuracy and resilience.
- Applied a data preprocessing pipeline: missing value handling, normalization, feature engineering, and cross-validation with hyperparameter tuning.
- Evaluated model performance using Accuracy, Precision, Recall, F1-Score, and AUC-ROC in a 3-member team; managed feature selection and model training.

Student Record Management System | HTML, MySQL

- Developed a web-based CRUD application to store, retrieve, and manage student academic records using MySQL and HTML forms.
- Implemented search and filter functionality to query records by roll number, class, and result status.

RPA Automation Workflow | AutomationEdge RPA

- Designed and deployed an RPA workflow during AutomationEdge Foundation training to automate a repetitive data-entry task.
- Reduced simulated manual processing time by automating form filling and data transfer steps between applications.

CERTIFICATIONS

- Introduction to Databases - Meta / Coursera, Feb 2025 (coursera.org/verify/1GHPYC011PEJ)
- AutomationEdge Foundation (RPA) - AutomationEdge, Sep 2025 (Cert No: 20756C3R6BOI1Z4)
- Coding and Programming - Samsung Innovation Campus, Nov 2025 - Mar 2026, Telecom Sector Skill Council
- Red Hat Application Development I: Programming in Java EE (AD183) - Red Hat, Jun 2025, 32 Credit Hours
- Red Hat OpenStack Administration I: Core Operations for Domain Operators (CL110) - Red Hat, Mar 2026, 40 Credit Hours
- Software Developer & QA Tester Internship - ShieldGuardia Software Private Limited, May 2026 - Jul 2026 (Cert No: SG-2026-008)

CAREER OBJECTIVE

Seeking a long-term role in Software Development, Machine Learning, or Data Engineering - leveraging research experience in NLP, time-series forecasting, and bio-inspired optimization, along with practical skills in Python, MySQL, and RPA - within a technology-driven organization where I can contribute meaningfully and grow as a CS professional.